

My capstone project will expand on my machine learning course project, which focuses on predicting film box-office revenue. This broader study will explore how machine learning can serve as a decision-support tool for film investment. In the current version of the project, I use movie genres, keywords, and selected metadata to forecast box-office revenue using supervised learning models. The results indicate that both text-based and structured content signals provide valuable predictive insights.

However, the current project has several significant limitations that I plan to address in my capstone. First, although numerical metadata features such as budget, runtime, and release timing were prepared during data processing, they were not fully integrated into the final modeling pipeline. Second, the model faces challenges with highly skewed revenue distributions and the presence of major blockbuster outliers. Lastly, the evaluation method relies on random train/test splitting, which does not accurately reflect real-world forecasting conditions. Instead, a time-aware split would be more appropriate for this context.

Hence, for my capstone project, I plan to address several existing gaps in the following three ways. First, I will gather additional data and develop a more comprehensive feature set that includes information about the cast and director, early signals from critics or audiences, and richer textual features beyond just keywords. Second, I will explore more robust multimodal models that can learn from text, categorical data, and numerical inputs simultaneously, rather than primarily relying on TF-IDF representations. Third, I will redesign the evaluation framework to incorporate temporal splitting and introduce metrics relevant to investment, allowing for a more realistic assessment of the model's effectiveness in decision-making contexts. The final deliverable will be a capstone presentation that effectively communicates the project's motivation, methodology, results, and practical implications for film investment.